

Experimenting with Google Gemini: A Multimodal Model

Estimated time: 30 minutes

Welcome to the lab, Experimenting with Google Gemini: A Multimodal Model.

Note: Google Gemini may not be available in all countries, so this exercise is optional.

Learning objectives

After completing this lab, you will be able to:

- Navigate Google AI Studio.
- Explore the Gemini Pro Vision generative AI model.
- Save the results for future use.

Introduction

Google AI Studio is a web-based tool that gives users free access to Google's generative AI models, such as Gemini, Gemini Pro, and Gemini Pro Vision. Users can prompt the models, experiment with different scenarios and use cases, and export code in many popular programming languages, including Python and JavaScript.

What is Gemini AI?

Gemini replaces PaLM 2, the Generative AI model behind the chatbot Google Gemini. It is built with multimodality in mind, which means that it is capable of understanding text, images, videos, audio, and code. Gemini is the first model to outperform human experts on MMLU (Massive Multitask Language Understanding), one of the most popular methods to test the knowledge and problem-solving abilities of AI models.

In this lab, you'll explore Google AI Studio with a focus on using the Gemini Pro Vision model to:

- Select an image
- Generate a prompt based on the image selected
- Save the results to your drive
- Export the results as code

Exercise 1: Log into Google AI Studio

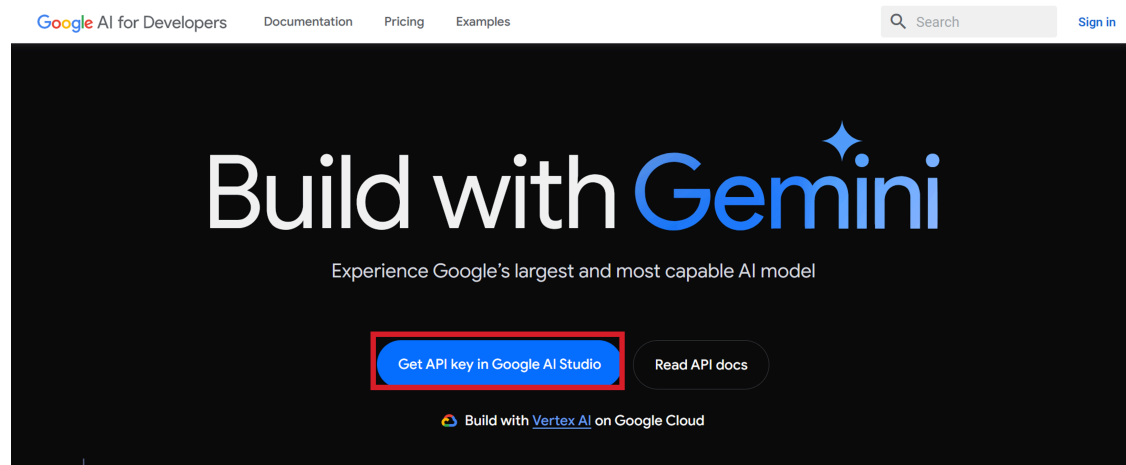
Let's get started with how you can use Gemini Pro Vision.

1. Visit the [Build with Gemini](#) webpage.

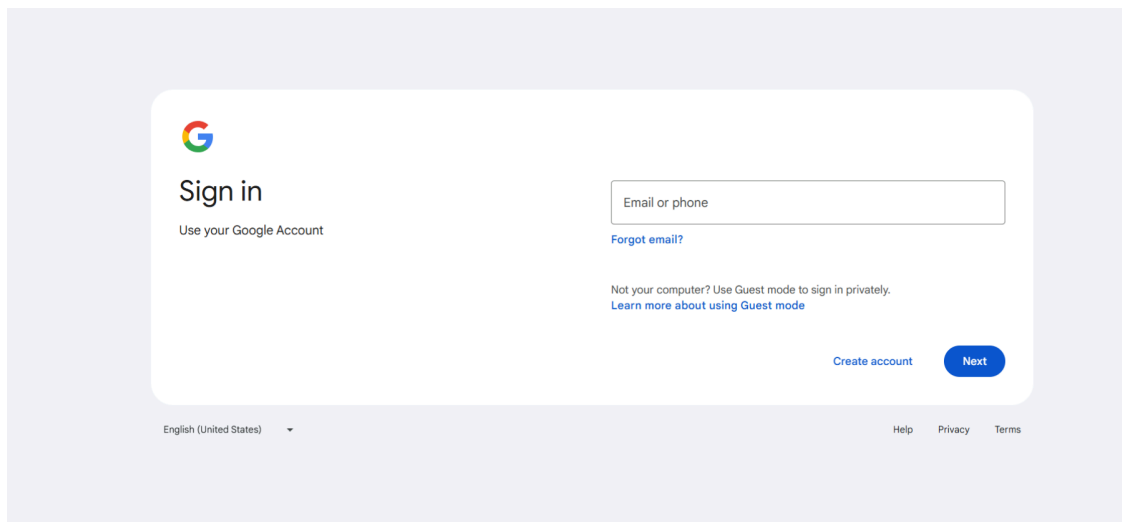
Note: If you cannot access the webpage by clicking on the URL, copy the following URL and paste it in a browser.

<https://ai.google.dev/>

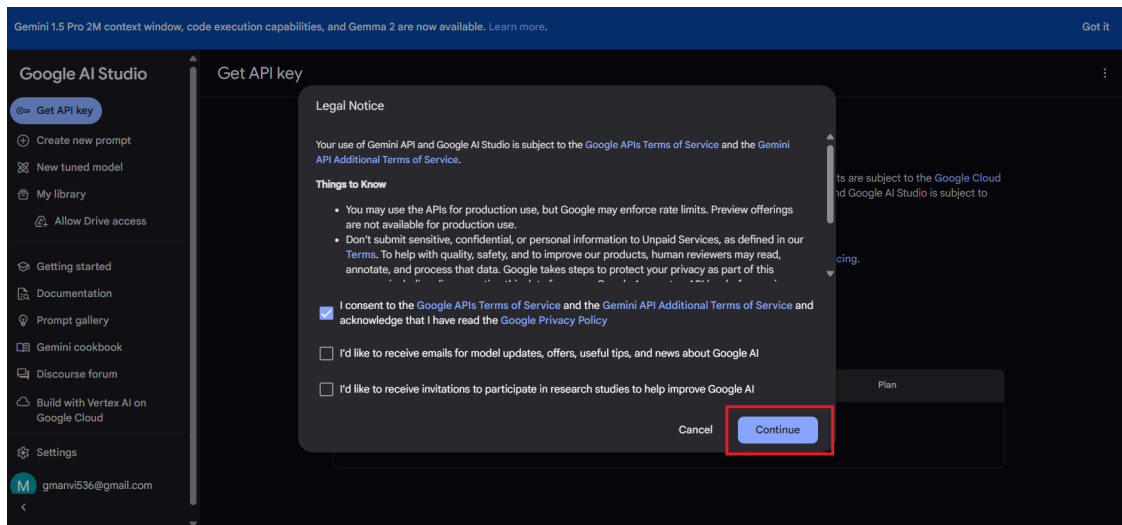
2. Click on **Get API key in Google AI Studio**.



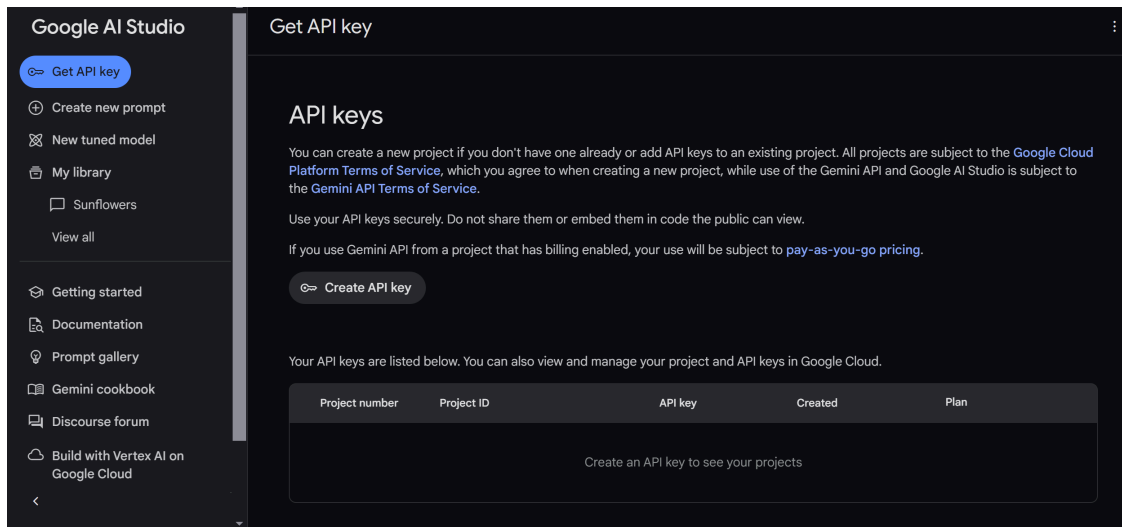
3. You will be redirected to the Google Sign-in page. Sign in with your Google account or set up a Google account if you do not have one.



4. After successfully signing in, you will reach the **Google AI Studio APP** page. However, you must first read and consent to the Terms of Service to proceed.



5. You have reached the Google AI Studio APP page.

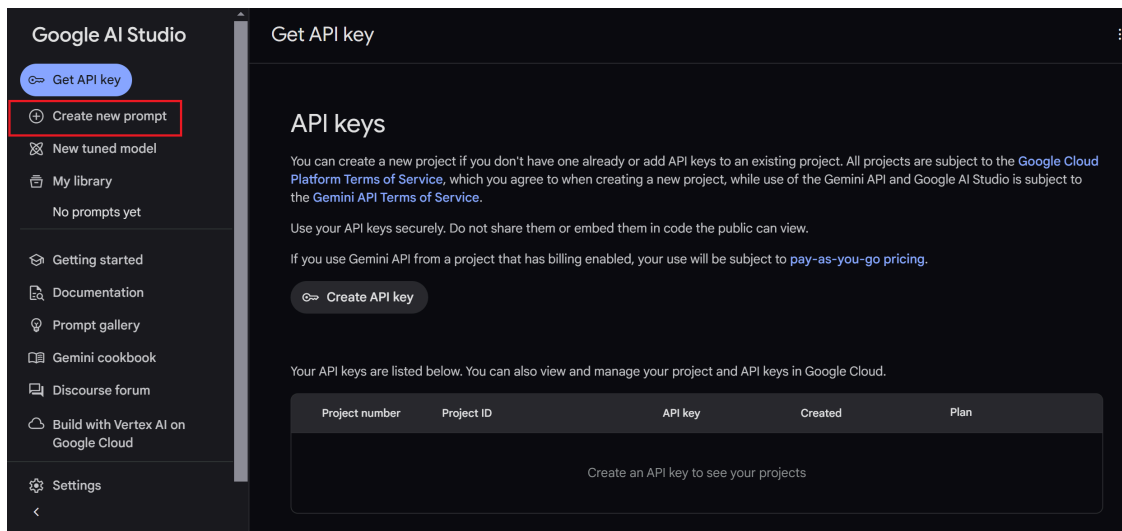


Exercise 2: Build a prompt

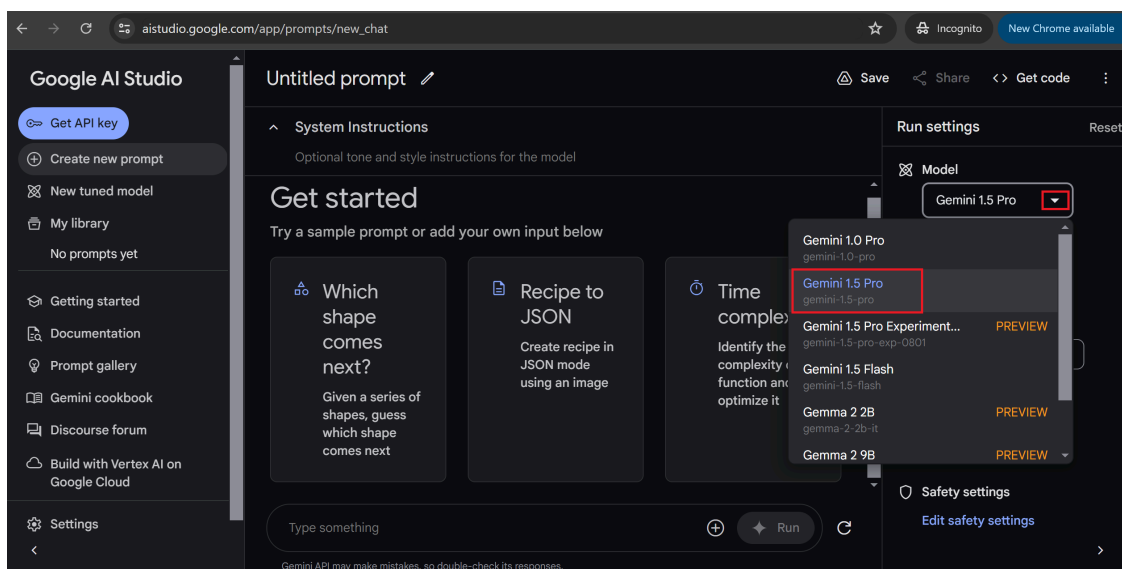
Gemini's multimodal abilities let you prompt the model with a combination of imagery and text. For example, you can use this feature to learn more about specific features of an image. Let's try to create a prompt with text and images.

Step 1: Create a multimodal prompt

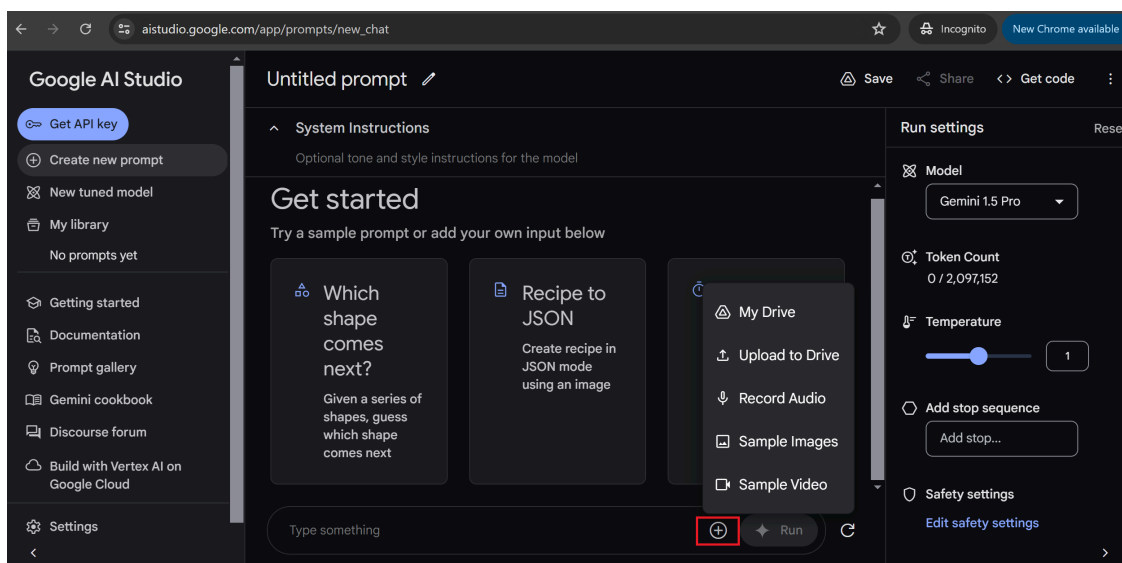
1. In the left panel, click on Create new prompt as highlighted in the screenshot below.



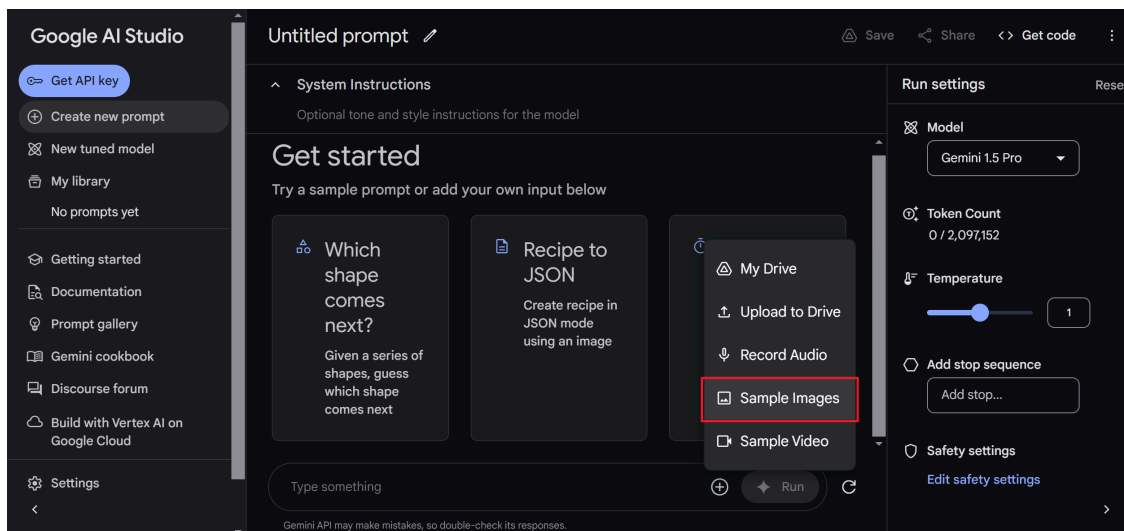
2. In the right column **Model** field, select a model that supports images, such as the **Gemini 1.5 Pro** model.



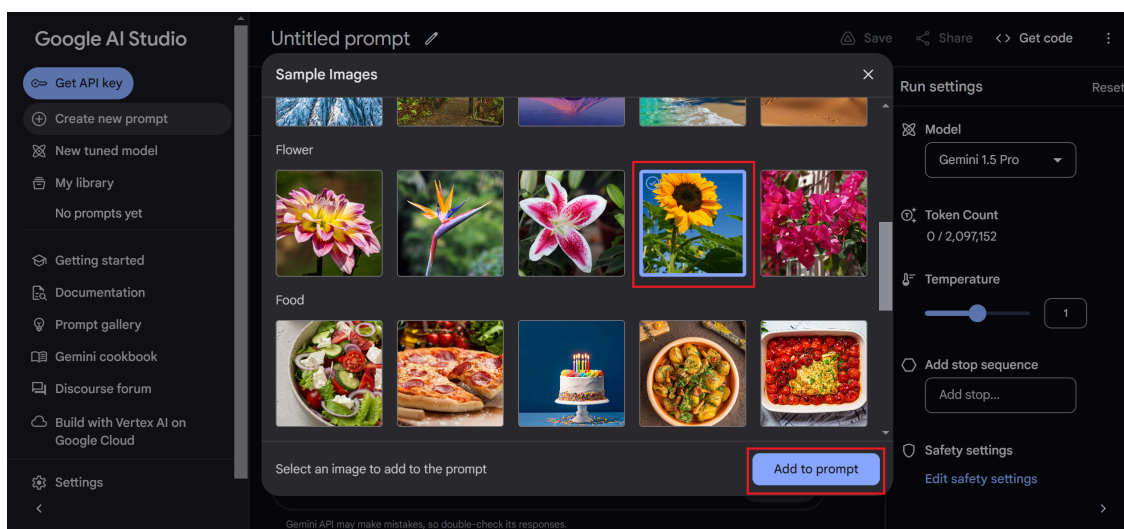
3. Next, click on the "+" icon, where you will be able to see multiple prompt options here.



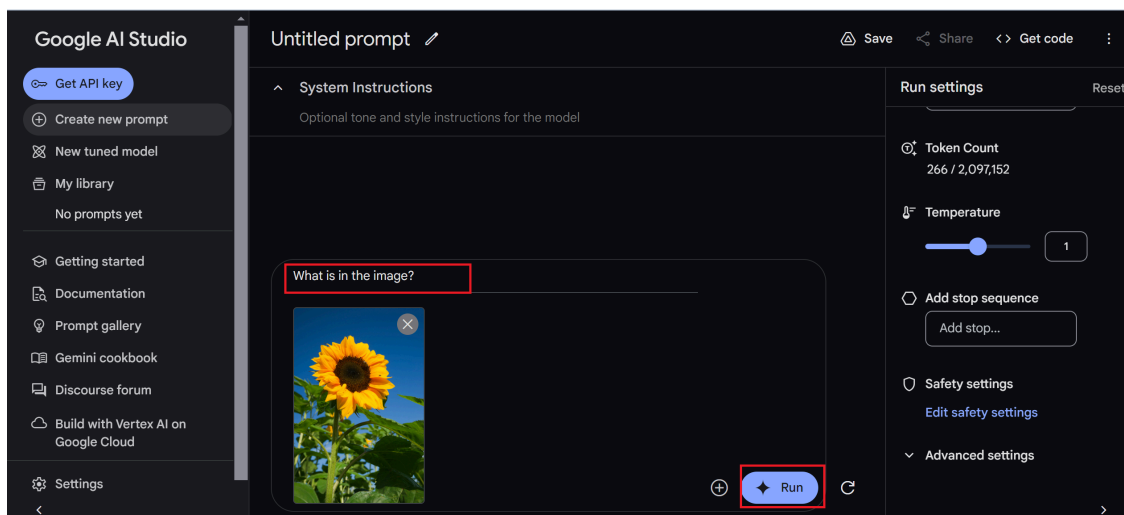
4. From the prompt area, select Sample Images.



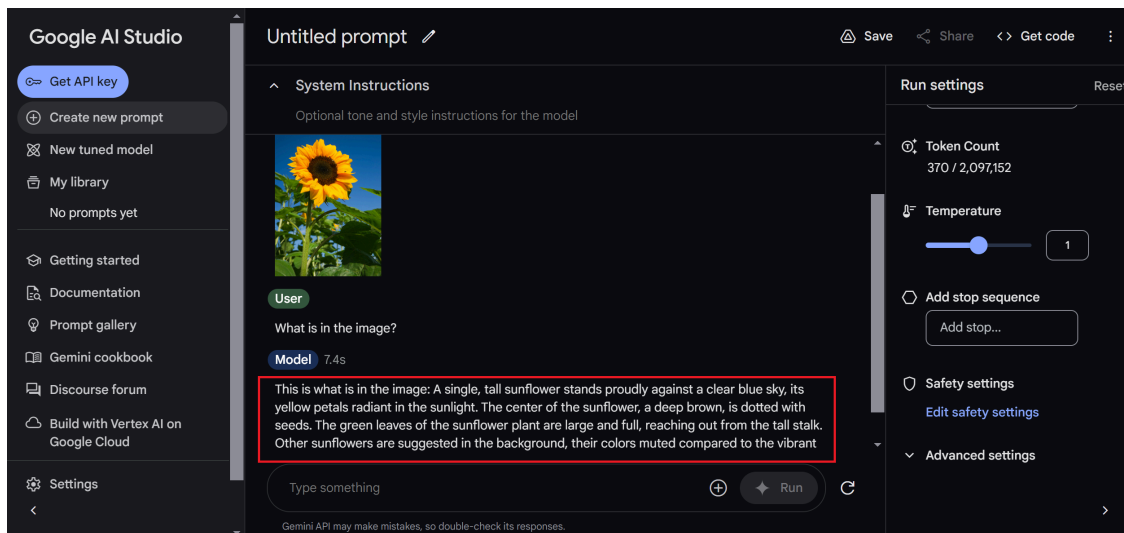
5. Choose one of the Sample image and click on **Add to prompt** button.



6. Enter your text in the **Type something** field.



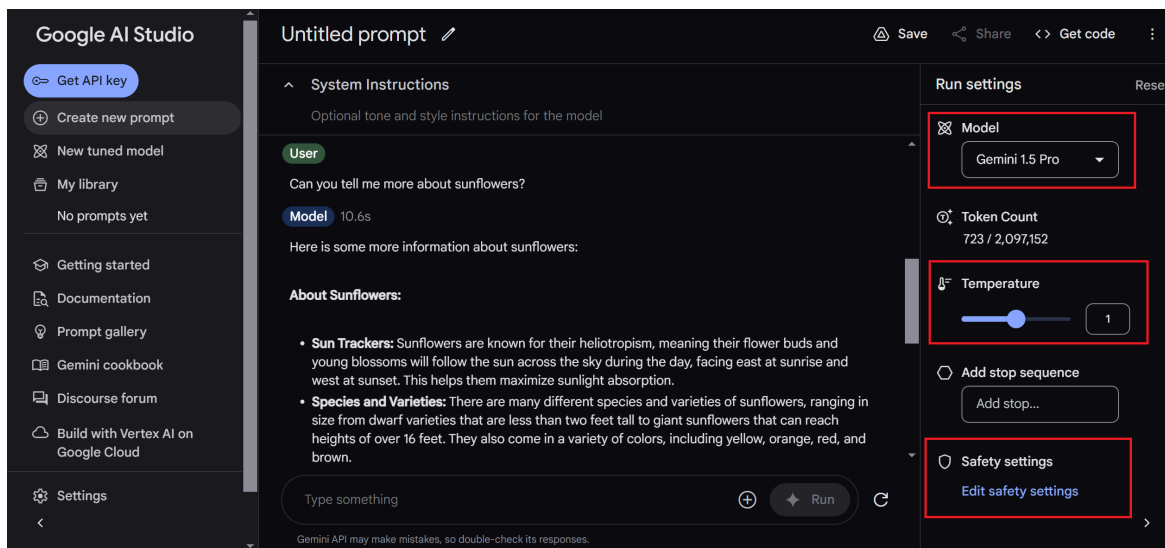
7. Click the **Run** button to create a description of the selected image.



You have successfully prompted the model with a fixed string of text and an image.

Good-to-know information: Experiment with model parameters

As you're prototyping your prompt, you can also play around with model run settings on the right side of the application. These are key settings to know about.



Model

Select what model you want to respond to your prompts. For more information about the available models and capabilities, see [Models](#).

Temperature

Control how much randomness is allowed in the model's responses. Raising this value allows the model to produce more unexpected and creative responses.

Safety settings

Adjust safety settings for managing model responses. For more details about these controls, see the [Safety settings](#).

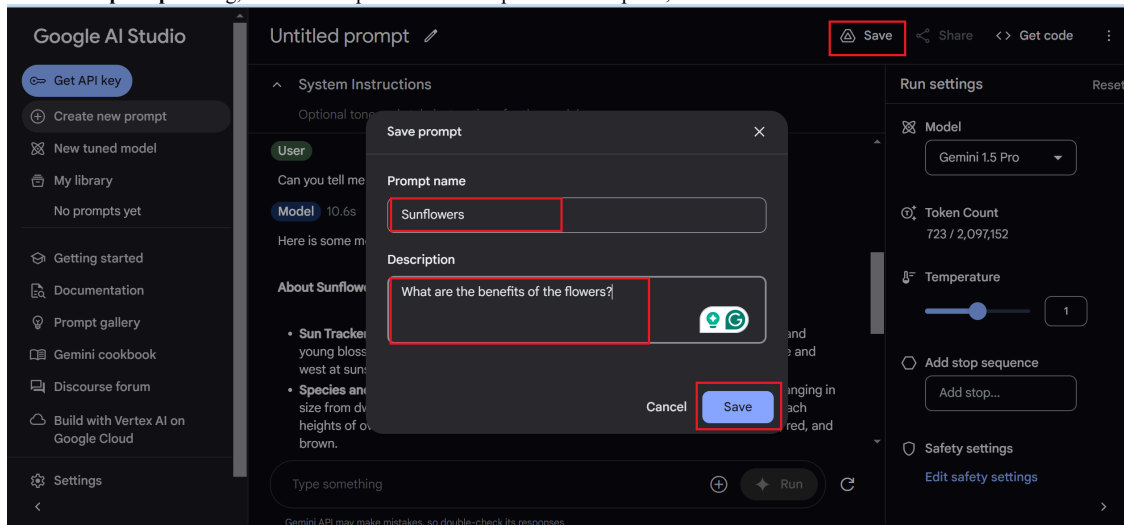
Save or export your prompt

Now that you've prototyped a generative AI application, you can save your work or generate code to use this prompt in your own development environment.

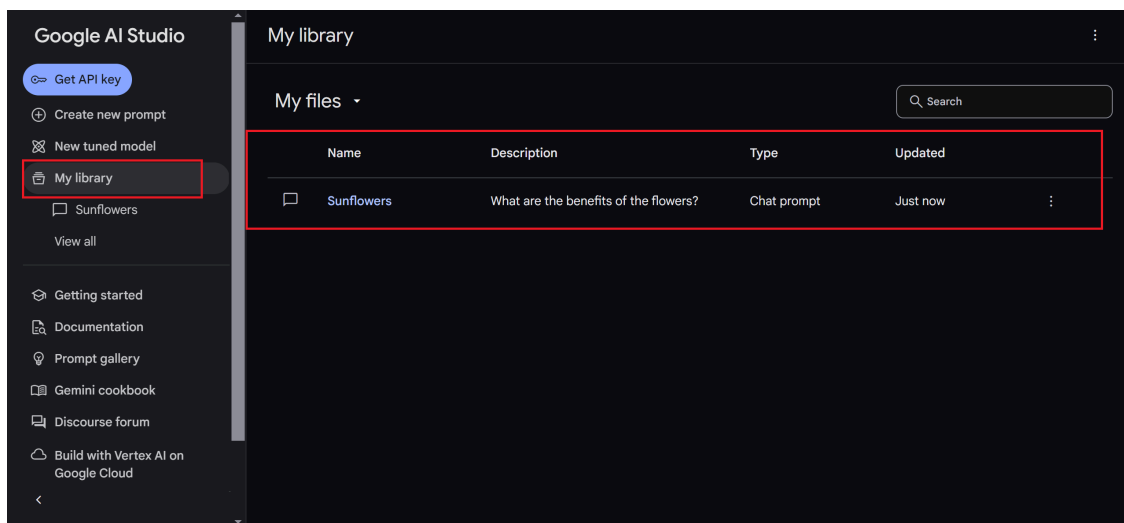
To save the prompt you created:

1. In the top right corner of the Google AI Studio app, select **Save**.
2. Connect the app to your Google Drive account, if you want.

3. In the **Save prompt** dialog, enter a Prompt name and an optional Description, and then select **Save**.



NOTE: You can save the prompt to your Google Drive or save it to **My library** in the Google AI Studio.



[Build with Gemini](#)

Summary

Congratulations on completing the hands-on lab, Experimenting with Google Gemini: A Multimodal Model.

Using an image as a starting point to generate a prompt is an interesting experiment. There are many other features of the Google AI Studio that you can explore at your own pace. For instance, working with Gemini Pro and text prompts.

Visit the Google DeepMind webpage to learn more - <https://deepmind.google/technologies/gemini/#>



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